

Image Stitching / Image Mosaicing Notes

1 Overview

Image stitching (or mosaicing) combines multiple overlapping images into a single large image. The typical pipeline is:

1. Detect feature points (SIFT)
2. Compute feature descriptors
3. Match features between images
4. Estimate homography
5. Use RANSAC to remove outliers
6. Warp images and blend them

2 SIFT (Scale Invariant Feature Transform)

SIFT detects distinctive keypoints that are invariant to scale, rotation, and illumination changes.

Main steps:

2.1 Scale Space Construction

The scale space is built using Gaussian blurring.

$$L(x, y, \sigma) = G(x, y, \sigma) * I(x, y)$$

where

- $I(x, y)$ = input image
- $G(x, y, \sigma)$ = Gaussian kernel

2.2 Difference of Gaussians

Keypoints are detected using the Difference of Gaussians (DoG):

$$D(x, y, \sigma) = L(x, y, k\sigma) - L(x, y, \sigma)$$

Local extrema in this scale-space are potential keypoints.

2.3 Keypoint Localization

Unstable points are removed by eliminating:

- low contrast points
- edge responses

2.4 Orientation Assignment

Each keypoint is assigned a dominant orientation based on image gradients.

Gradient magnitude:

$$m(x, y) = \sqrt{(L_x)^2 + (L_y)^2}$$

Gradient orientation:

$$\theta(x, y) = \tan^{-1} \left(\frac{L_y}{L_x} \right)$$

2.5 Descriptor Generation

A 16×16 neighborhood is divided into 4×4 cells.

Each cell stores an 8-bin orientation histogram.

Descriptor dimension:

$$4 \times 4 \times 8 = 128$$

3 Feature Matching

Descriptors from two images are compared using Euclidean distance.

Nearest neighbor matching is performed.

Lowe's ratio test:

$$\frac{d_1}{d_2} < 0.75$$

This removes ambiguous matches.

4 Homography

Homography describes a projective transformation between two images.

$$x' = Hx$$

where

$$H = \begin{bmatrix} h_1 & h_2 & h_3 \\ h_4 & h_5 & h_6 \\ h_7 & h_8 & h_9 \end{bmatrix}$$

Points are represented in homogeneous coordinates:

$$x = (x, y, 1)^T$$

5 Homography Estimation

From matched points we construct a system:

$$Ah = 0$$

Each point correspondence provides two equations.

Minimum correspondences required:

$$4$$

6 Solving Using SVD

We compute the Singular Value Decomposition:

$$A = U\Sigma V^T$$

The solution h is the singular vector corresponding to the smallest singular value.

$$h = v_{last}$$

The vector h is reshaped into the 3×3 homography matrix.

7 RANSAC

Feature matching produces outliers.

RANSAC (Random Sample Consensus) estimates a robust homography.

Algorithm:

1. Randomly select 4 matches
2. Compute homography
3. Count inliers
4. Repeat for many iterations
5. Select model with maximum inliers

8 Image Warping

After estimating H , we warp one image to align with the other.

$$x' = Hx$$

Inverse warping is commonly used to avoid holes in the output image.

9 Image Blending

To remove seams between images:

- feathering
- multiband blending
- exposure compensation

10 Applications

- panorama generation
- satellite image mosaics
- medical imaging
- 3D reconstruction